Project Title: Traffic Management System

Objectives:

- Monitor real-time traffic conditions at strategic locations.
- Collect data on traffic flow, density, and speed.
- Provide actionable insights for traffic management and city planning.

1. Description:

Give the title and description to reflect that this is a traffic management project.

2. Hardware Setup:

In the Wokwi workspace, use the IoT board to a suitable board for traffic management, such as "NodeMCU" or "ESP8266."

3. Connect New Sensors:

Connect the new traffic sensors to the IoT board as required. Ensure that you follow the wiring instructions for the specific sensors you are using.

4. Code:

The code to accommodate the new sensors and the requirements of a traffic management system. For example,to develop a Python script on the IoT devices to send real-time traffic data to the traffic information platform

```python
import paho.mqtt.client as mqtt
import json
import time
from random import randint

# Simulating traffic data
def generate_traffic_data():
    return {
        'location': 'Intersection A',
```

```python
        'timestamp': int(time.time()),
        'traffic_level': randint(1, 5)  # Random traffic level for 
demonstration
    }

# MQTT settings
mqtt_broker = 'your_broker_address'
mqtt_topic = 'traffic_data_topic'

# Callbacks
def on_connect(client, userdata, flags, rc):
    print(f"Connected with result code {rc}")
    client.subscribe(mqtt_topic)

def on_publish(client, userdata, mid):
    print(f"Message {mid} published.")

# Create MQTT client
client = mqtt.Client()
client.on_connect = on_connect
client.on_publish = on_publish

# Connect to broker
client.connect(mqtt_broker, 1883, 60)

# Main loop
try:
    while True:
        traffic_data = generate_traffic_data()
        payload = json.dumps(traffic_data)

        # Publish data to the topic
        client.publish(mqtt_topic, payload)

        # Sleep for a short interval (e.g., 1 second)
        time.sleep(1)

except KeyboardInterrupt:
    print("Script terminated by user.")
    client.disconnect()
```

5. Simulate the Traffic Management System:

Click the "Run Simulation" button to test your modified code with the new sensors.

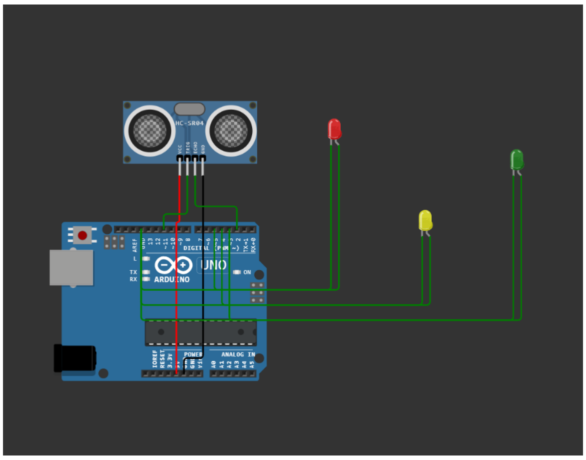
Ensure that the IoT board simulates the traffic management tasks and provides real-time data related to traffic conditions.

6. Monitor and Debug:

Use the Wokwi Serial Monitor to observe and debug your code .

Process further when there is no any issues.

7. Simulated Image:



The IoT Traffic Monitoring System offers a technological solution to the challenges of urban traffic, providing decision-makers with valuable insights for better traffic management and fostering a data-driven approach to city planning. The project represents a step towards smart cities that leverage IoT technologies to create more efficient and sustainable urban environments.